slopgs — MIDI Implementation Chart

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This document describes the MIDI behavior of the `slopgs` MIDI implementation, an emulation of the Microsoft GS Wavetable Synth.

Updates along with `SPEC.adoc`.

1. Basic Specifications

Number of Parts

16.

Maximum Polyphony

54 voices (48 sustained under continuous overload)

Effects

None.

Wavetable source

A DLS-1 sample collection. (Depicted here is the standard `gm.dls`.)

Output format

22050 Hz, stereo, 16-bit signed PCM.

2. Receive data

In every byte pattern below, `n` is the MIDI channel nibble (`0`–`F`, `0`=channel 1 … `F`=channel 16).

2.1. Channel Voice Message

`8n kk vv` — Note off

Releases the note on its patch's own release curve, at the same rate in decibels per second no matter how far the note had already decayed — so a note released late fades out sooner, not more slowly.

- `kk` = note number (`00`–`7F`, 0–127)
- `vv` = release velocity (`00`–`7F`) — **ignored**

`9n kk vv` — Note on

- `kk` = note number (0–127)
- `vv` = velocity (`01`–`7F`, 1–127) - fixed curve. Perceptually, amplitude scales with the square of velocity/127

  Velocity is not only a volume control: on patches whose sample collection asks for it, a harder-struck note also **attacks** faster. In `gm.dls` 27 of the 235 instruments do. Note number behaves the same way for decay — see §4.2.

`An kk vv` — Polyphonic Key Pressure (aftertouch)

Not recognized. Received, no audible effect.

`Bn cc vv` — Control Change

See §2.2.

`Cn pp` — Program Change

- `pp` = program number (`00`–`7F`, 0–127)

  Combines with whatever Bank Select value is currently in effect (§2.2) to select an instrument.

`Dn vv` — Channel Pressure (aftertouch)

Not recognized. Received, no audible effect.

`En ll mm` — Pitch Bend Change

- `ll` = LSB
- `mm` = MSB
- Combine to a 14-bit value `mm` × 128 + `ll`.
- Center is `00 40` (8192, no bend)
- Applied continuously to every sounding note on the channel — a bend arriving mid-note is heard **immediately**, not just on the next note. Range is set by RPN0, default ±2 semitones (§2.2).

2.2. Control Change

Only the controllers below are recognized. **Every other CC number is received and produces no effect, silently — never an error.**

`Bn 00 mm` — Bank Select MSB

See Bank Select LSB.

`Bn 20 ll` — Bank Select LSB

GS Mode only: Takes effect on the **next Program Change**, never on notes already sounding.

> **WARNING**
>
> GS Mode CCs are **dropped outright** if GS mode is off — the value is not stored, not queued, not remembered.
>
> GS mode is entered only by a GS Reset System Exclusive (§2.6), and is left again by GM System On, GM System Off or System Reset (`FF`)

`Bn 01 vv` — Modulation Wheel

Recognized.

`Bn 06 mm` — Data Entry MSB

See Data Entry LSB.

`Bn 26 ll` — Data Entry LSB

Recognized only together with RPN 0/1/2 selection, below; meaningless on their own.

`Bn 07 vv` — Channel Volume

Recognized. Default `64` (100).

`Bn 0A vv` — Pan

Recognized. Default `40` (64, center).

`Bn 0B vv` — Expression

Recognized. Default `7F` (127).

`Bn 40 vv` — Hold 1 (Sustain)

Recognized.

- `00` = off (pedal up) — default
- `01`–`7F` = on (pedal down). **Any non-zero value**, not just `40`.

`Bn 5B vv` — Effects 1 Depth (Reverb Send)

Not recognized. Received w/o effect.

`Bn 5D vv` — Effects 3 Depth (Chorus Send)

Not recognized. Received w/o effect.

`Bn 62 ll` — NRPN LSB

See NRPN MSB.

`Bn 63 mm` — NRPN MSB

Recognized only as "deselect the current RPN". **No NRPN parameter is ever addressable** — Data Entry sent after an NRPN selection has no effect.

Sending either, with any value, immediately clears whatever RPN was previously selected to Null.

`Bn 64 ll` — RPN LSB

See RPN MSB.

`Bn 65 mm` — RPN MSB

Selects a Registered Parameter for the following Data Entry (CC6/CC38) messages to target:

RPN (MSB,LSB)	Parameter	Applied
`00 00`	**Pitch Bend Sensitivity.** **Data MSB only**: Semitones (default=2). **Not range-checked** — any MSB `00`–`7F` is accepted. Bends are still capped at ±48 semitones. LSB ignored.	Live — read continuously, alongside Pitch Bend itself.
`01 00`	**Channel Fine Tuning** 14-bit Data Entry: −100 … +99 cents.	**Latched at Note On.** Changing it while a note is held does not retune that note.
`02 00`	**Channel Coarse Tuning** **Data MSB only**: −64 … +63 semitones. LSB ignored.	**Latched at Note On**, same as Fine Tuning.
`7F 7F`	**Null** (no parameter selected).	—
Others	**Any**	Stored, but **no effect**

`Bn 78 00` — All Sound Off

Recognized. Releases (does not hard-cut) every sounding note on the channel, **bypassing** a held sustain pedal.

`Bn 79 00` — Reset All Controllers

Recognized Bank/Program and the RPN/NRPN selection are left untouched, otherwise re-arms:

Parameter	Value
Channel Volume	100
Pan	64 (center)
Expression	127
Pitch Bend	8192 (center)
Modulation	0
Sustain pedal	Released

> **WARNING**
>
> Reset All Controllers (CC121), GS Reset, GM System On/Off and System Reset (`FF`) **DO NOT reset** Pitch Bend Sensitivity (RPN0), nor which RPN is currently selected — both survive every reset path.
>
> Fine Tuning (RPN1) and Coarse Tuning (RPN2) are **not** in that exception: CC121 leaves them alone, but GS Reset, GM System On/Off and System Reset (`FF`) all zero them.

`Bn 7A 00` — Local Control

Not applicable. Received, no effect.

`Bn 7B 00` — All Notes Off

Recognized. Releases every sounding note on the channel, **honoring** a held sustain pedal (notes under a held pedal stay held).

`Bn 7C 00` — Omni Mode Off

See Omni Mode On.

`Bn 7D 00` — Omni Mode On

Not recognized. Received, no mode change (§2.3).

`Bn 7E mm` — Mono Mode On (Poly Off)

Recognized. Switches the channel to mono and **releases** every currently-sounding note on it (implicit all-notes-off, per the MIDI spec), bypassing a held sustain pedal exactly as All Sound Off does.

`mm` (requested mono voice count) is **ignored** — the value sent changes nothing.

`Bn 7F 00` — Poly Mode On

Recognized. Switches the channel to poly and **releases** every currently-sounding note on it (implicit all-notes-off), bypassing a held sustain pedal.

2.3. Channel Mode Message

Mode is only ever **Omni Off** (each channel answers exactly the Part(s) mapped to it — see Basic Channel in the chart, §3). Omni On/Off messages are received but never change this.

Mono/Poly are switchable per-channel via CC126/CC127 (§2.2); a channel is otherwise Poly by default.

2.4. System Common Message

Song Position Pointer, Song Select and Tune Request: **not recognized.** `slopgs` renders a Standard MIDI File on its own clock and has no use for sequencer-sync messages.

2.5. System Real Time Message

`F8` Timing Clock, `FA` Start, `FB` Continue, `FC` Stop

Not recognized.

`FE` Active Sensing

Not recognized. Received, no effect.

`FF` Reset

Recognized. Restores the power-on defaults below. Note that it does **not** restore everything — see the warning after the table.

Parameter	Value
Bank/Program	0 / 0 / 0
Channel Volume	100
Pan	64 (center)
Expression	127
Pitch Bend	8192 (center)
Modulation	0
Sustain pedal	Released
Mono/Poly	Poly
Fine / Coarse Tuning (RPN1 / RPN2)	0 / 0
Data Entry	0
Master Volume	Unity
Per-Part tuning grid	Flat (0)
Part→channel map	Back to default (see §3.1)
Rhythm parts	Part `0` only
GS mode	Cleared (Bank Select inert until the next GS Reset)

WARNING

The currently-selected **RPN/NRPN parameter** and the **Pitch Bend Sensitivity (RPN0)** value survive a Reset unchanged.

Fine/Coarse Tuning (RPN1/RPN2) do not — a Reset zeroes both.

2.6. System Exclusive Message

Four SysEx envelopes are recognized. The first is Roland's `DT1` ("Data set 1"), which carries one parameter address per message and is expanded in §2.6.1 below; the other three are the standard Universal messages.

`F0 41 dev 42 12 aa bb cc dd…dd sum F7`	Roland GS-family message. • `dev` = device ID — **not checked** • `aa bb cc` = 21-bit parameter address. • `dd⋯dd` = data bytes for that address. • `sum` = checksum — **not checked**
`F0 7E dev 09 01 F7`	GM System On (Universal Non-Realtime). Full device reset, and **clears GS mode** — Bank Select goes inert again until the next GS Reset. `dev` not checked.
`F0 7E dev 09 02 F7`	GM System Off (Universal Non-Realtime). Full device reset, and **clears GS mode** — exactly as GM System On does. `dev` not checked.
`F0 7F dev 04 01 ll mm F7`	Master Volume (Universal Realtime). `dev` not checked. **Only `mm` (MSB) is used; `ll` (LSB) is never read.**

2.6.1. Roland Messages

The addresses listed below are DT1 addresses, to be filled in the `aa bb cc` part of the Roland GS-family message above.

Address	Parameter	Effect
`40 00 7F`	GS Reset	Full device reset, **and** sets GS mode.
`40 1n 02`	RCV CHANNEL (GS Mode)	Remaps which incoming channel Part `n` listens to. Persists until a Reset.
`40 1n 15`	USE RHYTHM PART (GS Mode)	Nonzero = Part `n` switches to drum-kit ("rhythm") mode. **Also fires a Reset All Controllers on every channel** — all notes cut, controllers back to defaults. Bank/Program survive.
`40 1n 40–4B`	Scale Tuning One signed offset per semitone class (C…B), centred on `40`, stored per Part `n`	Recognized. Twelve data bytes are consumed, always starting at semitone class C.
`40 1n` *any other*	Anything else in Roland's Patch-parameter address space for Part `n` (Reverb/Chorus Macro, Part Level, Pan, Pitch Key Shift, Velocity Sense, …)	**Not individually implemented.**

NOTE

Only `40 00 7F` (GS Reset) works from a cold start. Everything else is **ignored unless GS Mode is activated via said Reset.**

Not recognized:

- System Parameters (Master Tune, Master Volume-via-`DT1`, Master Key-Shift, Master Pan — address family `40 00 00–06`)
- Display Data (address family `10 00 xx`, Model ID `45`)
- Drum Setup Parameters (address family `41 xx xx` — per-drum-instrument key/level/pan/reverb/chorus edits)
- Bulk Dump (address families `48 xx xx`/`49 xx xx`)

Any `DT1` (or `RQ1`) message with an address outside the table above, or with a manufacturer ID other than `41`/`7E`/`7F`, is accepted and silently dropped.

Fallthrough `40 1n xx..` messages ("not individually implemented") are instead consumed by the same 12-byte write that serves Scale Tuning, regardless of which parameter the address nominally names, and always starting at semitone class C. **A sequence intended to set, e.g., Reverb Macro or Part Level on real GS hardware will overwrite that Part's whole Scale Tuning table instead.**

3. MIDI Implementation Chart

Basic Channel

The MIDI channel each of the 16 Parts answers to. Parts are numbered `0`–`15`, matching the `n` in the `40 1n xx` System Exclusive addresses (§2.6).

Mode

Refers to the MIDI Mode messages (Omni On/Off, Mono/Poly). `slopgs` always behaves as Omni Off; only the Mono/Poly half of Mode is switchable, per channel.

Note Number / True Voice

The range of note numbers accepted, and the range actually reproduced. Note number 60 is middle C.

Velocity

Whether Note On and Note Off messages' velocity byte affects anything.

After Touch

`Key’s` = polyphonic (per-note) aftertouch, `Ch’s` = channel (monophonic) aftertouch.

Control Change

See §2.2 for the full per-controller list — this row just points there.

Program Change

The program numbers a Program Change message can select (raw message value — one less than the corresponding instrument number as usually printed on an instrument list).

System Exclusive

See §2.6 for the full address table.

System Common, System Real Time

Sequencer/drum-machine sync messages. Not used, except Reset (`FF`).

3.1. Implementation Chart

Func	Transmit	Accept	Remarks
Basic Channel Default	×	1 – 16	Not an identity map. See the default parts mapping (§3.3) below.
Basic Channel Changed	×	1 – 16, each	Via GS RCV CHANNEL — which is itself inert until a GS Reset.
Mode Default	×	Mode 3	Omni Off, Poly.
Mode Messages	×	Mono, Poly (CC126/127)	Omni On/Off received, no effect.
Mode Altered	×	Mode 3, 4	
Note Number	×	0 – 127	Depends on the loaded `gm.dls`.
True Voice	×	0 – 127	Depends on the loaded `gm.dls`.
Pitch Bender	×	O	14-bit, continuous.
Control Change	×	see below	Its own table, directly below this one.
Prog Change	×	O	
True #	×	0 – 127	
System Exclusive	×	O	• GM On/Off • GS Reset • RCV CHANNEL • USE RHYTHM PART • Master Volume • Scale Tuning • **Device ID and checksum unchecked**
System Common	×	×	
System Real Time	×	×	Except Reset — see Aux Messages, below.
Aux Messages			
Local ON/OFF	×	×	
All Notes OFF	×	O (CC123)	
Active Sense	×	×	
Reset	×	O (`FF`)	
Velocity			
Note ON	×	O	
Note OFF	×	×	
After Touch			
Key’s	×	×	
Ch’s	×	×	

3.2. Control Change

CC	Transmit	Accept	Remarks
Bank Select 0 / 32	×	cond.	No effect until a GS Reset has been received.
Mod Wheel 1	×	O	
Data Entry 6 / 38	×	O	Only meaningful with RPN, below.
Channel Volume 7	×	O	Default 100.
Pan 10	×	O	Default 64 (center).
Expression 11	×	O	Default 127.
Hold 1 64	×	O	
Reverb Send 91	×	×	No reverb processor exists.
Chorus Send 93	×	×	No chorus processor exists.
NRPN 98 / 99	×	cond.	Only deselects the current RPN; no NRPN parameter is ever addressable.
RPN 100 / 101	×	O	Pitch Bend Sensitivity / Fine Tune / Coarse Tune only.
All Sound Off 120	×	O	Bypasses a held sustain pedal.

CC	Transmit	Accept	Remarks
Reset All Ctrl 121	×	O	Bank/Program and RPN selection untouched.
Local Control 122	×	×	No local keyboard to disconnect.
All Notes Off 123	×	O	Honors a held sustain pedal.
Omni Off 124	×	×	
Omni On 125	×	×	
Mono On 126	×	O	Also releases the channel (implicit all-notes-off, bypasses the pedal).
Poly On 127	×	O	Also releases the channel (implicit all-notes-off, bypasses the pedal).
Other —	×	×	No effect, no error.

> **NOTE**
>
> - `O` : recognized/handled.
> - `x` : received (or n/a) with no effect.
> - `cond.` : recognized, but only under a stated condition — see Remarks.
>
> Mode 1: Omni On, Poly.
> Mode 2: Omni On, Mono.
> Mode 3: Omni Off, Poly.
> Mode 4: Omni Off, Mono.

3.3. Default Parts Mapping

On boot, the parts-channels mapping are set as follows:

Part	Channel	Assignment
0	10	Rhythm
1	1	Melodic
2	2	Melodic
3	3	Melodic
4	4	Melodic
5	5	Melodic
6	6	Melodic
7	7	Melodic
8	8	Melodic
9	9	Melodic
10	11	Melodic
11	12	Melodic
12	13	Melodic
13	14	Melodic
14	15	Melodic
15	16	Melodic

These mappings can be changed via the Roland SysEx RCV CHANNEL and USE RHYTHM PART messages (§2.6.1).

4. Miscellaneous Quirks

4.1. Polyphony

An automatic accelerated-release policy engages from the **49th simultaneous voice** — not at 54 — freeing voices faster than a plain steal-oldest policy would.

A voice cut this way, or cut by a drum kit's exclusive-key group, reaches silence within 14 ms of the cut however long its patch's own release is; it fades, never clicks off.

4.2. Envelopes

Amplitude fades at a constant rate in decibels, so a note fades at the same slope whatever level it is released from, and a patch's authored decay and release times mean "the time to fall 96 dB". Both are also **scaled per note** by the sample collection: higher notes decay faster, and on some patches harder-struck notes attack faster.

5. Presets

> **NOTE**
>
> All tables in this section depicts Microsoft's `gm.dls`, `slopgs` does not provide one. Your chosen DLS file will vary.

5.1. Instruments

#	Name (bank 0)	Variants (bank MSB: name)
		Piano
0	Piano 1	8: Piano 1 16: Piano 1d
1	Piano 2	8: Piano 2
2	Piano 3	8: Piano 3
3	Honky-tonk	8: Honky-tonk
4	E.Piano 1	8: Detuned EP 1 16: E.Piano 1v 24: 60's E.Piano
5	E.Piano 2	8: Detuned EP 2 16: E.Piano 2v
6	Harpsichord	8: Coupled Hps. 16: Harpsichord 24: Harpsi.o
7	Clav.	—
		Chromatic Percussion
8	Celesta	—
9	Glockenspiel	—
10	Music Box	—
11	Vibraphone	8: Vibraphone
12	Marimba	8: Marimba
13	Xylophone	—
14	Tubular-bell	8: Church Bell 9: Carillon
15	Santur	—
		Organ
16	Organ 1	8: Detuned Or.1 16: 60's Organ 1 32: Organ 4
17	Organ 2	8: Detuned Or.2 32: Organ 5
18	Organ 3	—
19	Church Org.1	8: Church Org.2 16: Church Org.3
20	Reed Organ	—

#	Name (bank 0)	Variants (bank MSB: name)
21	Accordion Fr	8: Accordion It
22	Harmonica	—
23	Bandoneon	—
	Guitar	
24	Nylon-str.Gt	8: Ukulele 16: Nylon Gt.o 32: Nylon Gt.2
25	Steel-str.Gt	8: 12-str.Gt 16: Mandolin
26	Jazz Gt.	8: Hawaiian Gt.
27	Clean Gt.	8: Chorus Gt.
28	Muted Gt.	8: Funk Gt. 16: Funk Gt.2
29	Overdrive Gt	—
30	DistortionGt	8: Feedback Gt.
31	Gt.Harmonics	8: Gt. Feedback
	Bass	
32	Acoustic Bs.	—
33	Fingered Bs.	—
34	Picked Bs.	—
35	Fretless Bs.	—
36	Slap Bass 1	—
37	Slap Bass 2	—
38	Synth Bass 1	1: SynthBass101 8: Synth Bass 3
39	Synth Bass 2	8: Synth Bass 4 16: Rubber Bass
	Strings	
40	Violin	8: Slow Violin
41	Viola	—
42	Cello	—
43	Contrabass	—
44	Tremolo Str	—
45	PizzicatoStr	—
46	Harp	—
47	Timpani	—
	Ensemble	
48	Strings	8: Orchestra
49	Slow Strings	—
50	Syn.Strings1	8: Syn.Strings3
51	Syn.Strings2	—
52	Choir Aahs	32: Choir Aahs 2
53	Voice Oohs	—
54	SynVox	—
55	OrchestraHit	—
	Brass	
56	Trumpet	—
57	Trombone	1: Trombone 2
58	Tuba	—
59	MutedTrumpet	—
60	French Horns	1: Fr.Horn 2

#	Name (bank 0)	Variants (bank MSB: name)
61	Brass 1	8: Brass 2
62	Synth Brass1	8: Synth Brass3 16: AnalogBrass1
63	Synth Brass2	8: Synth Brass4 16: AnalogBrass2
	Reed	
64	Soprano Sax	—
65	Alto Sax	—
66	Tenor Sax	—
67	Baritone Sax	—
68	Oboe	—
69	English Horn	—
70	Bassoon	—
71	Clarinet	—
	Pipe	
72	Piccolo	—
73	Flute	—
74	Recorder	—
75	Pan Flute	—
76	Bottle Blow	—
77	Shakuhachi	—
78	Whistle	—
79	Ocarina	—
	Synth Lead	
80	Square Wave	1: Square 8: Sine Wave
81	Saw Wave	1: Saw 8: Doctor Solo
82	Syn.Calliope	—
83	Chiffer Lead	—
84	Charang	—
85	Solo Vox	—
86	5th Saw Wave	—
87	Bass & Lead	—
	Synth Pad	
88	Fantasia	—
89	Warm Pad	—
90	Polysynth	—
91	Space Voice	—
92	Bowed Glass	—
93	Metal Pad	—
94	Halo Pad	—
95	Sweep Pad	—
	Synth Effects	
96	Ice Rain	—
97	Soundtrack	—
98	Crystal	1: Syn Mallet
99	Atmosphere	—
100	Brightness	—
101	Goblin	—

#	Name (bank 0)	Variants (bank MSB: name)
102	Echo Drops	1: Echo Bell 2: Echo Pan
103	Star Theme	—
	Ethnic	
104	Sitar	1: Sitar 2
105	Banjo	—
106	Shamisen	—
107	Koto	8: Taisho Koto
108	Kalimba	—
109	Bagpipe	—
110	Fiddle	—
111	Shanai	—
	Percussive	
112	Tinkle Bell	—
113	Agogo	—
114	Steel Drums	—
115	Woodblock	8: Castanets
116	Taiko	8: Concert BD
117	Melo. Tom 1	8: Melo. Tom 2
118	Synth Drum	8: 808 Tom 9: Elec Perc.
119	Reverse Cym.	—
	Sound Effects	
120	Gt.FretNoise	1: Gt.Cut Noise 2: String Slap
121	Breath Noise	1: Fl.Key Click
122	Seashore	1: Rain 2: Thunder 3: Wind 4: Stream 5: Bubble
123	Bird	1: Dog 2: Horse-Gallop 3: Bird 2
124	Telephone 1	1: Telephone 2 2: DoorCreaking 3: Door 4: Scratch 5: Wind Chimes
125	Helicopter	1: Car-Engine 2: Car-Stop 3: Car-Pass 4: Car-Crash 5: Siren 6: Train 7: Jetplane 8: Starship 9: Burst Noise
126	Applause	1: Laughing 2: Screaming 3: Punch 4: Heart Beat 5: Footsteps
127	Gun Shot	1: Machine Gun 2: Lasergun 3: Explosion

5.2. Drums

Kits (use Program Change):

Program	Kit
0	Standard
8	Room
16	Power
24	Electronic
25	TR-808
32	Jazz
40	Brush
48	Orchestra
56	SFX

Notes, incl. GM+GS:

Note	Sound	Note	Sound
27	High Q	58	Vibraslap
28	Slap	59	Ride Cymbal 2
29	Scratch Push	60	Hi Bongo
30	Scratch Pull	61	Low Bongo
31	Sticks	62	Mute Hi Conga
32	Square Click	63	Open Hi Conga
33	Metronome Click	64	Low Conga
34	Metronome Bell	65	High Timbale
35	Acoustic Bass Drum	66	Low Timbale
36	Bass Drum 1	67	High Agogo
37	Side Stick	68	Low Agogo
38	Acoustic Snare	69	Cabasa
39	Hand Clap	70	Maracas
40	Electric Snare	71	Short Whistle
41	Low Floor Tom	72	Long Whistle
42	Closed Hi-Hat	73	Short Guiro
43	High Floor Tom	74	Long Guiro
44	Pedal Hi-Hat	75	Claves
45	Low Tom	76	Hi Wood Block
46	Open Hi-Hat	77	Low Wood Block
47	Low-Mid Tom	78	Mute Cuica
48	Hi-Mid Tom	79	Open Cuica
49	Crash Cymbal 1	80	Mute Triangle
50	High Tom	81	Open Triangle
51	Ride Cymbal 1	82	Shaker
52	Chinese Cymbal	83	Jingle Bell
53	Ride Bell	84	Bell Tree
54	Tambourine	85	Castanets
55	Splash Cymbal	86	Mute Surdo
56	Cowbell	87	Open Surdo
57	Crash Cymbal 2		